When did plants first live on land? What the rocks can and cannot tell us

Barry W. Brook (University of Tasmania), with Claude Opus 5.5 (Anthropic) and OpenAI Codex. Draft of 8 November 2026; a plain-language summary of a living research repository, https://github.com/BWBrook/silurian_plants.

The puzzle

Every forest, grassland and crop descends from a few small plants that left the water hundreds of millions of years ago. Land plants changed the air, the soils and the climate, and built the stage on which animals came ashore. Yet scientists disagree by tens of millions of years about when the plants we know today began to diversify.

One way to date that is a *molecular clock*: compare the DNA of living plants and work out how long their lineages have been diverging. What it dates is the common ancestor of the living species, which need not be the first plant on land. Clocks need fossils to set the time scale, and they lean heavily on one kind of fossil argument: a *maximum*. A maximum needs a reason to think that a much older group would probably have left recognisable fossils in the rocks examined; how strong that reason is depends on where the earliest plants lived, what they produced, and how well their remains were preserved and searched for. Change the maximum and the clock's answer changes with it.

Three recent clock studies show how much rides on this. Two place the common ancestor of all living land plants 515–474 million years ago (Ma), in the Cambrian or earliest Ordovician. The newest, using a different set of fossil calibrations, places it at 457–450 Ma (Morris et al. 2018; Harris et al. 2022; Hooper et al. 2026). For the ancestor of living *vascular* plants, the group with internal plumbing that includes ferns, conifers and flowering plants, estimates run from about 452 to 428 Ma (figure below).

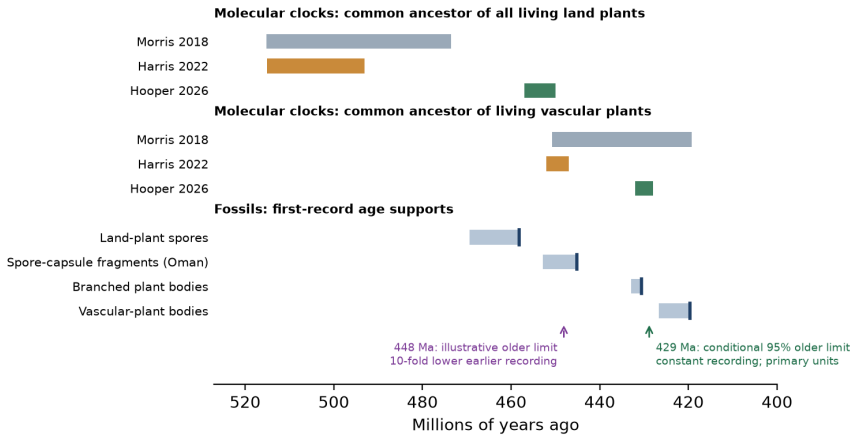


Figure. Clock estimates and first-record age supports (bands span earliest possible to earliest certain records; endpoints may be different finds). Arrows: limits on how far back the vascular-plant body record could reach if fossils were recorded at a steady rate (green) or ten times less often before about 427 Ma (purple). Neither is an origin date, and the fossils cannot choose between them.

Why the rocks are hard to read

The oldest traces of land plants are spores: tough, microscopic, and made in huge numbers. Dispersed land-plant spores are known from the Middle Ordovician. Late Ordovician rocks in Oman preserve fragments of spore-bearing capsules, and branched fertile plants, with stems and capsules often only millimetres long, are known from the Silurian. They record different parts of early plant biology, found in different ways, and do not measure how long a lineage existed before its first body fossil. So the absence of plant fossils from older rocks might mean the plants were not there, or only that small, woodless and perhaps rare plants were not preserved, found and recognised. A maximum has to argue that the first is likely.

What we did

For each maximum (515.5, 459.1 and 431.2 Ma) we asked what the fossil record can test, using three bodies of evidence.

- **Spores from Libyan boreholes.** In one North African basin, 28 rock samples older than about 443 Ma contain no spores bearing a three-rayed, or *trilete*, mark, while 51 of 59 younger samples do. Only 10 of the older samples may be older than 459 Ma, and those are broadly dated.
- **Spores from a Saudi Arabian borehole**, as a rougher check.
- **Dated fossils of early plant bodies**: 78 dated places, including one set of spore-capsule fragments from Oman, 50 of them with members of the vascular-plant group. From how these finds are spread through time, a standard method limits how far back a group's fossil record could reach; the limit depends on how often plants left fossils that were later found and recognised, so we tried several assumptions.

What we found

1. **The oldest maximum (515.5 Ma) rests on an argument the rocks we assessed cannot put numbers on.** Its authors argue that older marine rocks preserve tough algal cysts but no land-plant spores, so the spores' absence is real. Whether land spores reached those seas, and would have been found, cannot be measured here.
2. **The 459.1 Ma maximum gets one real test, and it is local and conditional.** In one Libyan basin, the samples rule out spore producers that were abundant through the later Ordovician and whose spores reached the rocks as well as later ones did. They cannot rule out rare producers, spores that did not reach those rocks, or plants that lived elsewhere. It allows a shared well effect, with samples otherwise independent.
3. **For vascular plants, the answer depends on assumptions the fossils cannot check.** At a steady recording rate, the conditional 95% older limit on their recognisable body record is about 429 Ma, or about 432–434 Ma if the finds are counted as fewer independent places. This limits the fossil record, not the origin date, and overlap with the youngest clock does not confirm that clock. If recognisable bodies appeared as soon as the group did, the older clocks' dates (444–452 Ma) would need such fossils to have been found 8 to 12 times less often before the oldest find than after it (2.4 to 4.5 if counted cautiously), or less if they appeared later.
4. **One fossil can move the limit by 20 million years.** Counting one set of tiny spore-capsule fragments from Late Ordovician Oman moves the calculated limit for land-plant bodies back by about 20 million years.

What it means

An older, sparsely recorded early flora is *allowed* by the evidence, but not *required* by it. An early lineage can leave few fossils because it was uncommon, local, hard to recognise or seldom preserved; our calculations show what that would allow, not which clock is right. Clock studies should state these assumptions and carry their uncertainty into the final dates.

How this work was done

Two AI systems, Claude (Anthropic) and Codex (OpenAI), did the research with Barry Brook, who set its direction; two independent reviews by GPT-6 Pro (OpenAI) shaped it. Every number traces to a file and script in the public repository.

Sources

Harris, Brogan J., James W. Clark, Dominik Schrempf, Gergely J. Szöllősi, Philip C. J. Donoghue, Alistair M. Hetherington, and Tom A. Williams. 2022. "Divergent Evolutionary Trajectories of Bryophytes and Tracheophytes from a Complex Common Ancestor of Land Plants." *Nature Ecology & Evolution* 6 (11): 1634–43. https://doi.org/10.1038/s41559-022-01885-x.

Hooper, Jack, Annabel M. Worth, Charles H. Wellman, James W. Clark, and Philip C. J. Donoghue. 2026. "Reconstructing Spore Evolution Recalibrates the Timescale of Land Plant Diversification." *Nature Plants*. https://doi.org/10.1038/s41477-026-02432-3.

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